

AD-CISPO: Token-Adaptive Clipped IS-Weight Policy Optimization

1 Thesis

MiniMax-M1 introduces CISPO, a policy-gradient objective that clips the importance-sampling (IS) weight rather than clipping away token updates as in PPO/GRPO-style surrogate objectives.¹ This proposal studies an adaptive extension, AD-CISPO, where the upper IS weight clipping radius is varied per token using frozen reference-model saliency. The goal is to keep CISPO’s dense token-gradient property while allocating a wider update corridor to tokens that appear structurally important for reasoning.

2 Notation

Symbol	Meaning	Shape or domain
q	prompt/question	text
o_i	response i in a sampled group	token sequence
$o_{i,t}$	token t of response i	token id
G	number of responses sampled per prompt	positive integer
π_θ	current policy being optimized	language model
$\pi_{\theta_{\text{old}}}$	rollout policy that produced the batch	language model
π_{ref}	frozen reference model for saliency and optional KL	language model
$r_{i,t}(\theta)$	IS weight $\frac{\pi_\theta(o_{i,t} q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, o_{i,<t})}$	positive scalar
$\hat{A}_i$	group-relative sequence advantage for response i	scalar
$S_{i,t}$	raw frozen-reference saliency for token $o_{i,t}$	nonnegative scalar
$m_{i,t}$	normalized token multiplier	nonnegative scalar, mean 1 over valid tokens
$\epsilon_{\text{low}}^{IS}$	lower IS clipping radius	nonnegative scalar
$\epsilon_{\text{high}}^{IS}$	base upper IS clipping radius	positive scalar
$\epsilon_{i,t,\text{high}}^{IS}$	token-adaptive upper IS clipping radius	positive scalar

3 MiniMax CISPO Baseline

The CISPO baseline starts from the off-policy REINFORCE objective with the IS weight treated as a stop-gradient coefficient:

$$J_{\text{ISPG}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(r_{i,t}(\theta)) \hat{A}_i \log \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (1)$$

¹<https://arxiv.org/html/2506.13585v1>

CISPO clips the IS weight itself:

$$\hat{r}_{i,t}(\theta) = \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{\text{high}}^{IS} \right), \quad (2)$$

and optimizes

$$J_{\text{CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (3)$$

The MiniMax-M1 paper reports that their CISPO experiments did not impose a lower IS-weight bound: they made $\epsilon_{\text{low}}^{IS}$ large and tuned only $\epsilon_{\text{high}}^{IS}$. The paper also states that CISPO uses no KL penalty term. A faithful MiniMax-style baseline should therefore set the KL coefficient to zero and either disable the lower bound or set it so low that it does not bind.

4 AD-CISPO Extension

AD-CISPO replaces the scalar upper clipping radius with a token-adaptive upper radius:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (4)$$

The clipped weight becomes

$$\hat{r}_{i,t}^{\text{AD}}(\theta) = \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{i,t,\text{high}}^{IS} \right), \quad (5)$$

and the objective is

$$J_{\text{AD-CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}^{\text{AD}}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (6)$$

The adaptive multiplier is computed from frozen-reference saliency:

$$\tilde{m}_{i,t} = \frac{S_{i,t}}{\frac{1}{|M_i|} \sum_{u \in M_i} S_{i,u}}, \quad (7)$$

where M_i is the valid generated-token mask. If all valid saliency values are zero or non-finite, $\tilde{m}_{i,t}$ falls back to 1 for valid tokens.

If lower or upper multiplier bounds are used, they must be followed by a renormalization or projection step so that the mean multiplier over valid tokens remains 1. A plain clamp such as $m_{i,t} = \max(\tilde{m}_{i,t}, \delta)$ is not sufficient because it changes the mean clipping budget.

5 Saliency Estimation

The main AD-CISPO implementation uses exact frozen-reference future-attention in-degree. For a selected reference layer ℓ and head h , let $a_{i,j,t}^{(\ell,h)}$ denote the causal attention probability from the query at token position j to the key/value position t . The saliency assigned to token $o_{i,t}$ is

$$S_{i,t} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \frac{1}{H_{\ell}} \sum_{h=1}^{H_{\ell}} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \quad (8)$$

Thus a token is salient when later valid generated tokens attend to it under the frozen reference model. The strict condition $j > t$ prevents self-attention from inflating saliency, and M_i excludes prompt positions and special-token sinks from both the source queries and target keys used for adaptive clipping.

The implementation computes this quantity blockwise from reference-model post-RoPE query and key states. This avoids materializing a full batch of $T \times T$ attention matrices while preserving the exact softmax attention weights for the selected layers, including the model’s causal, padding, and sliding-window masks.

A frozen-reference key-vector norm remains a cheaper ablation:

$$S_{i,t}^{\text{KV}} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \|k_{i,t}^{(\ell)}\|_2. \quad (9)$$

It should be reported as a proxy baseline, not as the primary AD-CISPO saliency definition.

6 Implementation Contract

A mathematically aligned AD-CISPO implementation should satisfy the following conditions:

1. CISPO loss must use a detached clipped IS weight multiplying $\log \pi_\theta$, not a PPO/GRPO surrogate minimum.
2. AD-CISPO must be enabled only with the CISPO loss, unless explicitly naming and evaluating an adaptive-GRPO ablation.
3. A MiniMax-faithful CISPO baseline should set KL weight to zero and avoid an active lower IS-weight bound.
4. Token-adaptive clipping should change only the upper IS clipping radius unless a lower-bound ablation is deliberately introduced.
5. The default AD-CISPO saliency should be exact frozen-reference future-attention in-degree. KV-norm saliency is only a lower-cost ablation.
6. Saliency tensors should be assigned to the generated token $o_{i,t}$ whose IS weight is being clipped. In an implementation where logits at position $t - 1$ predict token t , this corresponds to aligning generated-token saliency with the next-token log-probability terms rather than using the preceding hidden-state position.
7. Claims about compute and memory must distinguish active-policy overhead from frozen-reference overhead.

7 Evaluation Plan

The minimum evaluation should compare:

1. GRPO/PPO-style surrogate baseline.
2. MiniMax-faithful CISPO: no KL, no active lower IS bound, scalar upper IS radius.
3. AD-CISPO with exact future-attention in-degree saliency and token-adaptive upper radius.

4. AD-CISPO with KV-norm proxy saliency as a lower-cost ablation.
5. Ablation with shuffled or uniform multipliers to test whether saliency, rather than extra variance, causes any gain.
6. Optional adaptive-GRPO ablation, clearly separated from AD-CISPO.